

DAY 10 - SUMMARY

Functions

A function is a block of code that performs a specific task. Functions help reduce code repetition and improve program organization.

Types of Functions

1. Function Without Parameters and Without Return Type

Takes no input and returns no value.

Example:

```
void display() {  
    System.out.println("Hello");  
}
```

2. Function With Parameters and Without Return Type

- Takes input values but does not return a value.

Example:

```
void displayName(String name) {  
    System.out.println(name);  
}
```

3. Function Without Parameters and With Return Type

- Takes no input but returns a value.

Example:

```
int getNumber() {  
    return 10;  
}
```

4. Function With Parameters and With Return Type

- Takes input values and returns a result.

Example:

```
int add(int a, int b) {  
    return a + b;  
}
```

Types of Variables

Variables are used to store data in a program.

1. Local Variable

- Declared inside a method.
- Can be used only within that method.

Example:

```
void show() {  
    int age = 20; // Local Variable  
}
```

2. Instance Variable

- Declared inside a class but outside methods.
- Each object gets its own copy.

Example:

```
class Student {  
    String name; // Instance Variable  
}
```

3. Static Variable

- Declared using the static keyword.
- Shared by all objects of the class.

Example:

```
class Student {  
    static String college = "ABC College"; // Static Variable  
}
```